

RAS SEDR, Egypt

WMO: 624550

Lat: 29.5844N

Lon: 32.7175E

Elev: 53

StdP: 14.67

Time Zone: 2.00 (E02)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.6	49.6	26.4	20.4	60.2	30.5	24.7	61.0	21.7	61.3	17.9	62.7	3.9	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.0	100.9	71.2	99.0	71.1	97.1	71.1	77.2	89.3	76.1	88.2	75.3	87.3	8.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	126.7	83.2	72.7	121.7	82.7	71.7	117.3	82.4	40.7	89.8	39.7	88.2	38.8	87.1	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.1	13.7	DB	42.8	106.4	3.5	2.5	40.3	108.2	38.2	109.7	36.3	111.1	33.7	113.0	(4)
(5)			WB	37.4	80.2	2.7	2.3	35.5	81.9	33.9	83.2	32.4	84.5	30.5	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.0	58.7	61.3	66.1	71.9	78.7	84.1	87.2	87.5	83.1	77.1	69.3	62.1		
	DBStd	10.71	3.48	4.20	4.83	4.85	4.74	3.80	2.77	2.41	2.97	3.66	4.08	3.26		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	470	197	119	43	2	0	0	0	0	0	0	8	101		
	CDD50	8761	271	316	499	658	890	1024	1155	1162	994	839	579	374		
	CDD65	3757	3	15	78	209	425	574	690	697	544	374	137	11		
	CDH74	44039	7	71	464	1649	4543	7396	9662	9823	6576	3185	638	25		
	CDH80	22003	0	7	84	506	2063	4028	5540	5572	3121	998	84	0		
(14) Wind	WSAvg	6.8	5.3	5.7	7.1	7.6	7.7	8.1	7.5	7.8	7.8	6.9	5.5	5.1		
(15) Precipitation	PrecAvg	0.5	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecMax	1.1	0.5	0.3	0.9	0.1	0.3	0.0	0.0	0.0	0.0	0.2	0.4	0.3		
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.3	0.1	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	80.5	86.4	92.7	100.2	103.1	103.7	102.4	99.0	92.8	86.0	77.0		
		MCWB	60.1	61.1	62.7	63.1	67.7	69.4	71.8	72.4	70.4	71.1	66.0	62.4		
	2%	DB	70.9	75.8	82.0	88.6	95.6	99.7	100.9	100.0	95.7	89.5	82.4	73.5		
		MCWB	58.1	59.0	62.1	62.9	67.0	69.7	71.5	72.6	71.4	70.8	66.1	59.8		
	5%	DB	68.4	72.4	79.1	85.5	92.4	97.3	98.7	98.2	93.6	87.3	79.8	71.5		
		MCWB	56.1	57.5	60.6	62.4	66.3	69.5	71.4	72.4	71.7	70.2	65.4	58.8		
	10%	DB	66.6	69.9	76.0	82.5	89.7	94.8	96.7	96.1	91.6	85.2	77.6	69.8		
		MCWB	55.1	56.8	59.5	61.9	65.9	69.4	71.4	72.4	71.6	69.6	64.0	57.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	65.2	68.4	68.6	71.5	75.5	78.1	78.9	78.5	76.4	71.6	65.3		
		MCDB	71.4	73.6	77.8	82.6	88.9	89.7	92.4	90.9	87.7	85.2	79.9	73.3		
	2%	WB	59.8	61.2	64.5	66.3	69.8	73.9	76.4	77.4	75.9	74.4	69.4	62.2		
		MCDB	68.1	71.9	77.9	81.2	87.6	87.8	89.7	89.3	87.3	84.5	78.0	70.9		
	5%	WB	57.9	59.4	62.5	64.7	68.7	72.9	75.4	76.5	74.9	72.8	67.5	60.4		
		MCDB	66.8	70.0	75.3	79.7	86.0	87.3	87.8	88.4	86.3	82.7	76.8	69.5		
	10%	WB	56.1	57.8	60.8	63.4	67.6	71.8	74.5	75.6	73.9	71.5	65.6	58.8		
		MCDB	65.3	68.1	73.3	78.4	84.4	87.5	87.4	87.6	85.5	81.9	75.4	67.9		
(35) Mean Daily Temperature Range	MDBR		15.0	15.7	17.8	19.5	21.1	21.5	20.6	19.0	17.6	16.0	15.9	15.4		
	5% DB	MCDBR	17.1	19.2	21.4	22.9	23.9	23.8	22.2	20.7	19.6	17.9	17.2	17.1		
		MCWBR	11.2	11.0	10.6	9.5	8.4	7.7	7.1	6.4	6.9	7.8	9.7	10.8		
	5% WB	MCDBR	16.4	17.2	19.2	20.3	21.4	21.7	20.8	19.2	18.1	16.0	15.5	16.2		
		MCWBR	11.6	10.8	10.6	9.6	8.4	7.7	7.2	6.4	7.0	7.7	9.3	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.384	0.421	0.472	0.525	0.537	0.452	0.448	0.446	0.455	0.463	0.397	0.377		
	taud		2.165	2.045	1.894	1.786	1.784	2.032	2.067	2.078	2.044	2.012	2.178	2.207		
	Ebn at Noon		264	264	258	249	246	267	267	267	260	247	257	260		
	Edh at Noon		40	49	61	70	71	55	53	52	52	50	39	36		
(44) All-Sky Solar Radiation	RadAvg		1184	1481	1925	2279	2514	2676	2603	2415	2095	1671	1310	1109		
	RadStd		51	63	73	46	67	40	40	37	52	48	49	37		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (8 neighbors)	+0.97	N/A	-1.21	+1.04	N/A	N/A	-90	+382	+308	(47)

Nomenclature: See separate page